

Agentic AI Debate Coach

Presentation Analysis Platform

Final Project Documentation

Overview

A web platform that helps learners practise debates and improve presentations through role-based dashboards, argument analysis, coaching feedback, and performance reports.

Core capabilities

? Role-based dashboards for learners, educators, coaches, and administrators.

? Argument analysis with logical-fallacy detection, counterarguments, scoring, and feedback.

? Presentation analysis based on a transcript: clarity, structure, pacing, confidence, filler words, and transitions.

? Performance reports with skill breakdowns, activity visualizations, recommendations, and session transcripts.

Architecture

Frontend: React with Vite for dashboards, analysis tools, reports, and visualizations.

Backend: FastAPI and SQLAlchemy for authentication, REST APIs, and persisted history.

Analysis: deterministic presentation metrics alongside debate-analysis agents, giving actionable feedback even when an external AI service is unavailable.

Presentation Analysis Engine

Endpoint: POST /analysis/presentation

Input: a transcript of at least 20 characters and an optional duration from 15 seconds to 4 hours.

Output: overall score, four delivery metrics, word count, words per minute, filler-word count, transition count, and targeted recommendations.

Validation and quality assurance

Input validation is implemented with Pydantic. Unit tests check that scores remain in the 0-10 range and that filler words produce an actionable recommendation.

Verification completed: backend unit tests passed (2 tests) and the Vite production build passed.

Run locally

1. Backend: cd backend; activate the virtual environment; run `uvicorn app.main:app --reload`.
2. Frontend: run `npm install`, then `npm run dev`.
3. Open <http://localhost:5173> and select Presentation Analysis to submit a transcript.

Demo and presentation flow

1. Introduce the learner-feedback problem.
2. Show the role-based dashboard and debate tools.
3. Submit a presentation transcript and explain the generated delivery metrics.
4. Open the report to show visualized scores and coaching recommendations.

Screen recording

https://drive.google.com/file/d/15LqGuDEE7i2_hnmFg84qjLkdZUGtD2sz/view?usp=drive_web